
LogicWard

Live OT Drift-Detection & Incident-Response Platform

One detection fabric. Two live plants. Twelve real attacks.

A thermal power plant on a Raspberry Pi + a chemical reactor as a live 3D twin — monitored, attacked, detected and rolled back under one SOC console.

Adani OT Cybersecurity Demonstration

The problem: OT attacks are invisible until they hurt

- IT security assumes patchable, restartable, observable assets. **OT inverts every assumption.**
- Modbus has **no authentication**. A PLC cannot be patched mid-process; a “reboot” can trip the plant.
- The adversary’s goal is not data — it is **physical consequence**: defeat a safety interlock, then let physics do the damage.
- A lowered trip setpoint or an inverted safety comparator can sit **invisibly** in a running plant for weeks.

The gap

You cannot defend what you cannot see. Operators watch process values — not whether the *control logic itself* still matches what was approved.

Ukraine 2015/16, TRITON/TRISIS, and Oldsmar all began as unauthorized changes to trusted OT devices.

LogicWard: detect drift before it becomes risk

One sentence

LogicWard watches the **running control logic** and the **live process state**, and raises **severity-ranked, MITRE-mapped, evidence-backed** alerts the instant either drifts from a cryptographically signed, approved baseline.

What makes it credible

- Detection is a structural **diff**, not anomaly-guessing — **zero false positives**.
- The PLC program is **data to diff, never executed**.
- HMAC-signed baseline: tamper breaks the signature.

What makes it win

- **Multi-site, multi-vendor** — one fabric, two very different plants.
- A live **3D digital twin** of the chemical reactor.
- Production-grade SOC: risk, timeline, health, rollback.

Two plants, one detection fabric

Site A — Thermal Power Plant

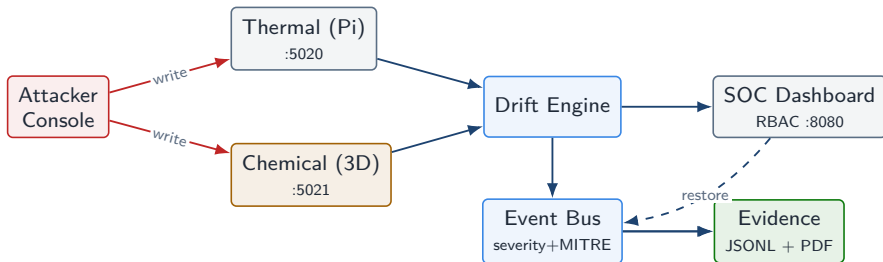
Raspberry Pi (real hardware)
Rockwell **L5X** ladder logic
Structural + register detection
Animated SVG SCADA mimic
thermal-pi ■ Modbus :5020

Site B — GRFICS Chemical Reactor

Live Unity 3D digital twin
OpenPLC **Structured Text**
Register-plane detection
Reactor + gauges + explosion
grfics-chem ■ Modbus :5021

The claim: not “I detect drift on my simulator” — but “I detect drift across *vendors, languages and processes* on one fabric.” A new plant is a *site profile*, not a rewrite.

Architecture: one enriched event bus is the spine



Every detector emits one Event. On emit the bus enriches it (**severity, MITRE, identity, sequence**) and fans out to the evidence log, the poll buffer and live subscribers. Events carry `details.site` — so two plants share one timeline.

Detection: six named PLC mutations

Mutation	Meaning	MITRE
condition stripping	safety input removed	T0889
logic inversion	comparator flipped	T0889
coil hijack	output repointed	T0889
rung injection	backdoor rung added	T0843
setpoint drift	threshold changed	T0836
register change	raw Modbus write	T0855

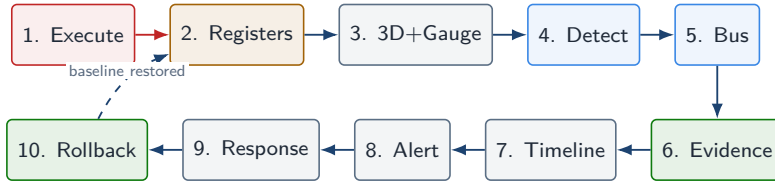
Two surfaces

Structural — canonicalize & diff the L5X rung-by-rung.

Register — diff live holdings/coils vs the signed snapshot.

A harmless re-export hashes identically
⇒ no alert. A real change is caught on the first 1-second scan.

Every attack runs the full lifecycle



On a live stage you can point to *every* stage on screen: the attacker fires, the registers move, the gauge/3D reacts, the alert appears with a MITRE badge, evidence is logged, and one click restores the approved baseline.

Twelve real attacks — over genuine channels

Site A — Thermal (7)

- Setpoint drift, force coil Modbus
- Logic inversion, condition stripping
- Coil hijack, rung injection Program
- DDoS flood T0814

Site B — Chemical (5, blast last)

1. Quality Sabotage stealth / colour
2. Feed Pump Trip level drains
3. Tank Overfill level rises + trip
4. E-Stop Injection plant halts
5. **Pressure Redline — BLAST**

Each chemical attack drives a **distinct** live signature — no two look alike on the gauges, detection feed or timeline.

The showpiece: a live 3D chemical reactor

- The real Fortiphyd **GRFICS Unity** reactor, embedded live in the SOC dashboard.
- Driven entirely by **live process data** — no scripted animation.
- The demo escalates: stealth sabotage → starvation → overfill → halt → **the reactor blasts**.
- Cinematic letterbox framing; the blast is **guaranteed** to fire (pressure to the ceiling in ~ 13 s).

Honest engineering

The Unity binary is pre-compiled; its process animations are what the binary maps. *All* per-attack distinctiveness is guaranteed on the live **SOC gauges + detection**, with a documented recompile path to unlock full 3D animation.

Production-grade SOC dashboard

- **Risk Score** — explainable 0–100 + band
- **Attack Timeline** — severity-coloured, site-filtered
- **System Health** — per-site status rollup
- **Rollback status** — integrity + drift + restore
- **Live trend sparklines** — rolling telemetry
- **Animated SCADA mimic** with pinned MITRE badges
- **Site selector** — Thermal / Chemical / All
- **Signed forensic PDF** — per site, on demand

Polling, not WebSockets — **deliberately**, for live-demo robustness.

Capability-based RBAC — six OT/ICS roles

Role	Scope	ack	ack all	baseline	net	safe	evidence
Operator	control room	✓	✓				
Control Engineer	PLC program	✓	✓	✓		✓	
Network Engineer	infrastructure	✓	✓		✓		
SOC Analyst	oversight	✓	✓		✓	✓	✓
Vendor / OEM	read-only						
CISO	full authority	✓	✓	✓	✓	✓	✓

Gated by **capability**, not rank — roles have different *scopes*. Enforced server-side (HTTP 403) *and* client-side. Vendor is read-only + monitored; CISO is full.

Verified, not hand-waved

154 automated checks

Nine integration suites spin up real servers on ephemeral ports and exercise real HTTP/Modbus paths — integration, not unit tests.

bus ■ I5x ■ plant	45
drift (6 mutations)	18
agent ■ dashboard (RBAC) ■ attacker	35
grfics (5 distinct chemical attacks)	36
multisite (unified feed, both sites)	20
Total	154

Every attack's **detection, severity, MITRE and site-tag** is asserted. MITRE IDs are **verified against the live ICS matrix**; where none fits, it is honestly N/A.

Why LogicWard wins the room

Differentiators

- **Multi-site, multi-vendor** on one fabric — rare in a demo.
- A **live 3D digital twin** that reacts to real attacks.
- **Explainable** detection + **verified** MITRE mapping.
- **Full lifecycle** on screen, ending in one-click rollback.

Enterprise-ready signals

- Capability RBAC for six real OT roles.
- Signed baseline + forensic PDF (audit trail).
- Risk score + health + timeline (SOC language).
- Process-agnostic — water / substation next.

Roadmap

- **Structured-Text program drift** for the chemical PLC — same six mutations, second parser.
- **Unity recompile** to hook liquid level / colour / e-stop strobes to the published JSON keys.
- **New site profiles** — water treatment, electrical substation — as drop-in plugins.
- **ML-assisted risk scoring** layered above the explainable rule-based core.

One detection fabric, many plants. **Research-grade. Demonstration-ready.**

LogicWard

Let's watch a reactor get attacked — and caught.

SOC dashboard :8080 ■ Attacker console :9090

Thank you.